

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and **COURSE CODE:** CSA1506
Big Data Analytics

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

Question Count: 5

Code of Conduct

I Ismail M (192524128) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

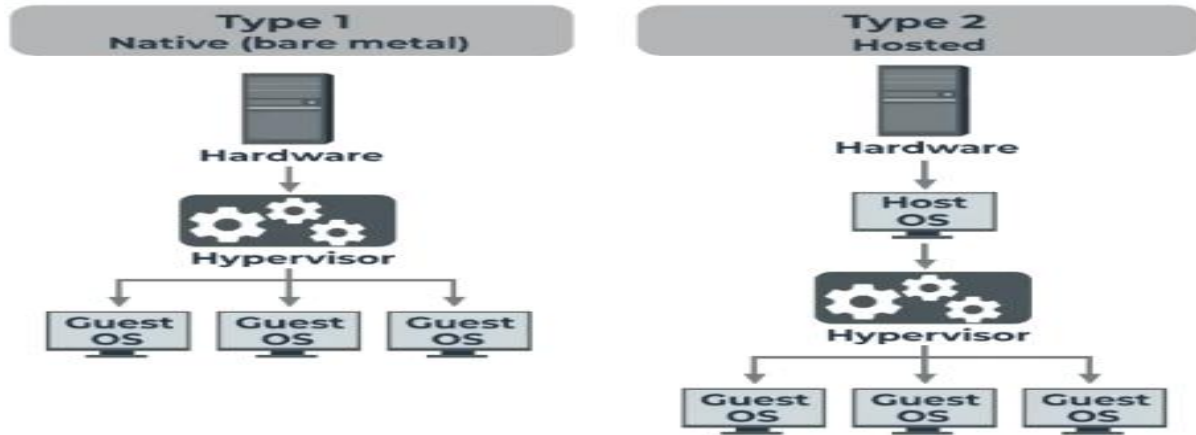
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ismail M

Assessment Tasks Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical application. Analyze performance, security, and manageability, then justify the better choice.

When hosting mission-critical applications (such as core financial transaction engines or enterprise ERPs), platform stability, zero-latency execution, and granular security controls are essential.



Analytical Comparison

Metric	Type-1 Hypervisor (Bare-Metal)	Type-2 Hypervisor (Hosted)
Architectural Layer	Installs directly on host bare-metal hardware. Examples: VMware ESXi, KVM, Microsoft Hyper-V.	Runs as an application layer on top of a conventional host operating system (OS). Examples: VirtualBox, VMware Workstation.
Performance	Near-Native Speed. Eliminates intermediate host OS context-switching overhead, offering low latency and direct hardware pass-through.	High Overhead. Emulated calls pass through guest OS → hypervisor → host OS → hardware, creating I/O and latency bottlenecks.

Metric	Type-1 Hypervisor (Bare-Metal)	Type-2 Hypervisor (Hosted)
Security	Minimal Attack Surface. Thin hypervisor kernel with no general-purpose host OS services or unneeded device drivers exposed to exploits.	Elevated Risk. Vulnerabilities in the underlying host OS directly compromise the hypervisor and all running guest VMs.
Manageability & HA	Built for enterprise clusters; supports live migration (e.g., Motion), automated fault tolerance, and dynamic load balancing (DRS).	Lacks enterprise cluster coordination, distributed resource management, and enterprise-grade SLA enforcement.

Architectural Recommendation & Justification

Recommendation: Type-1 (Bare-Metal) Hypervisor.

Justification: Mission-critical cloud workloads require strict Service Level Agreements (SLAs), low tail-latency, and continuous availability. A Type-1 hypervisor interacts directly with underlying CPU/RAM hardware registers without reliance on a host OS layer. This guarantees determinism, reduces the potential exploit surface area, and enables seamless live cluster failover during hardware maintenance or outages.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

When isolating high-compliance workloads—such as **Finance** (PCI-DSS compliance), **Healthcare** (HIPAA compliance), and **Student Portals** (public/less-trusted access)—on shared physical hardware:

How Virtualization Architecture Enforces Isolation

1. **Hardware-Assisted CPU Privileges:**

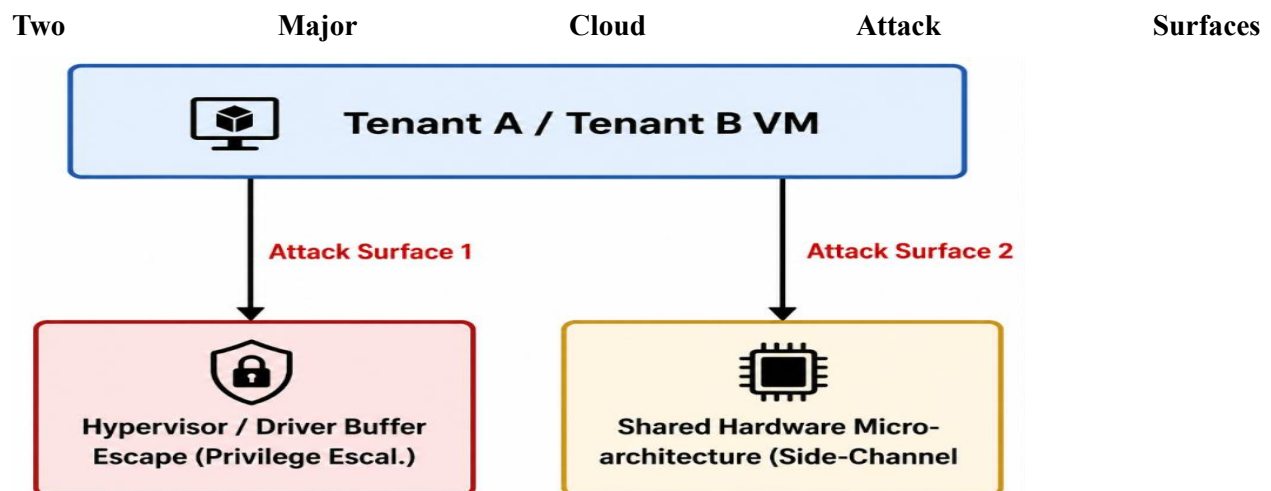
Modern processors use specialized execution modes (Intel VT-x / AMD-V) to enforce privilege boundaries. Guest operating systems operate in *VMX Non-Root Mode* (Ring 1/3), while the hypervisor retains *VMX Root Mode* (Ring 0). Guest VMs cannot execute privileged CPU instructions or access restricted system registers directly.

2. **Memory Virtualization via Hardware Tables:**

Hypervisors implement Extended Page Tables (EPT) or Nested Page Tables (NPT) to maintain isolated physical memory page mappings for each tenant. A VM's virtual memory space cannot address memory blocks allocated to another tenant's VM.

3. **Software-Defined Network Micro-Segmentation:**

Virtual switches (vSwitches) isolate tenant traffic using overlay encapsulation protocols (VXLAN / Geneve) and logical firewalls. This prevents cross-tenant packet sniffing or unauthorized inter-VM network access on the same hypervisor.



1. Hypervisor Escape / VM Breakout:

- **Mechanism:** An attacker in a low-privilege VM exploits a zero-day flaw in an emulated device driver (e.g., virtual network card or storage controller) managed by the hypervisor process.
- **Impact:** The attacker executes code directly within hypervisor memory (Ring 0), bypassing all software guest boundaries and gaining root access over all other tenant workloads on that physical host.

2. Microarchitectural Side-Channel Attacks (e.g., Spectre, Meltdown, Prime+Probe):

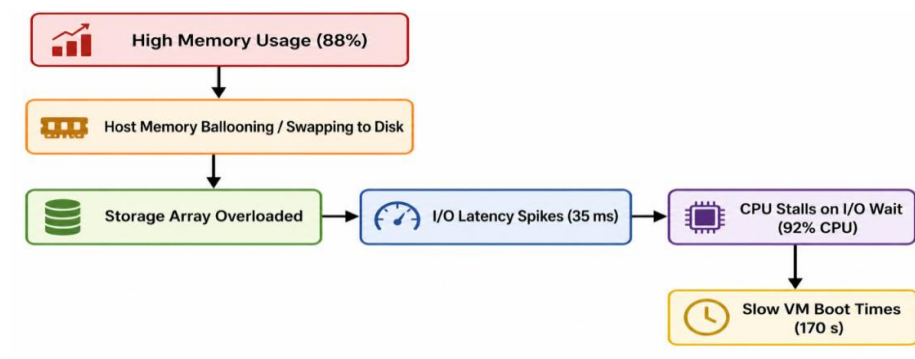
- **Mechanism:** Co-located VMs execute on shared physical CPU cores and share L1/L2/L3 hardware caches, execution pipelines, and branch predictors.
- **Impact:** A malicious tenant measures cache hit/miss timing differences to infer sensitive cryptographic keys or data residing in the CPU cache of a adjacent finance or healthcare VM on the same socket.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

System Snapshot Analysis

- **CPU Utilization:** 92% (Critical Saturation)
- **Memory Utilization:** 88% (High Contention Threshold)
- **Storage I/O Latency:** 35 ms (Severe Delay; healthy enterprise baseline is < 5 ms)
- **Average VM Boot Time:** 170 s (Extremely Slow; typical baseline is < 30 s)

Root Cause & Bottleneck Analysis



Primary Bottleneck — Storage I/O Subsystem: A 35 ms storage latency indicates severe disk queue depth exhaustion. During VM boots, intense random read operations hit saturated storage arrays, dragging out boot times to 170 seconds.

1. **Secondary Bottleneck — Memory Paging & CPU I/O Wait:** An 88% memory load causes the hypervisor to invoke host-level paging/swapping to disk. This extra read/write load further starves storage bandwidth. Consequently, the CPU reaches 92% utilization largely due to elevated **I/O Wait cycles** (iowait), where CPU cores sit idle waiting for storage operations to return.

Recommended Corrective Actions

Immediate / Operational Remediation

- **Live Migrate (vMotion) Workloads:** Immediately move 20–30% of non-critical VMs to less-utilized host nodes in the cluster to drop CPU and RAM pressure.
- **Apply Storage QoS:** Enforce maximum IOPS limits on high-volume background tasks to free queue depth for booting VMs.

Short-Term Optimization

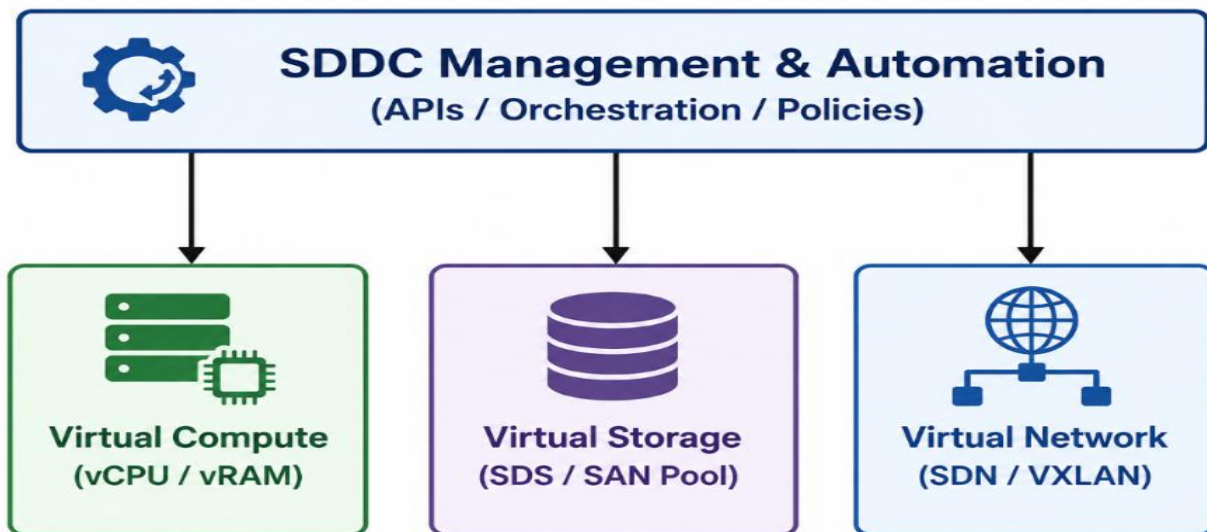
- **Disable Host Swapping:** Increase physical RAM allocation or enforce strict VM memory reservations to eliminate hypervisor swapping to storage.
- **Storage Tiering & Caching:** Move virtual machine disk images (.vmdk / .qcow2) and boot templates to high-speed NVMe flash tiers.

Long-Term Architectural Upgrades

- **Transition to All-Flash / NVMe-oF:** Replace legacy SAS/SATA SAN storage arrays with NVMe over Fabrics (NVMe-oF) storage architectures.
- **Cluster Scale-Out:** Add extra compute nodes to lower average cluster host memory utilization below 70% and CPU utilization below 65%.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Software-Defined Datacenters replace manual hardware configuration with automated, software-controlled resource management.



Agility Breakdown by Virtualization Tier

1. Compute Virtualization:

- **Traditional:** Provisioning physical servers requires hardware procurement, racking, power installation, and OS setup (taking days or weeks).
- **SDDC Agility:** Compute capacity is virtualized into abstract pools. Provisioning a virtual server takes seconds via automated API calls, with dynamic load balancing (DRS) placing workloads automatically.

2. Storage Virtualization (Software-Defined Storage - SDS):

- **Traditional:** Involves manual LUN masking, fabric zoning, RAID group creation, and physical SAN/NAS provisioning by dedicated administrators.
- **SDDC Agility:** SDS pools local NVMe/SATA disks across nodes into unified storage pools. Storage Performance Policies (e.g., IOPS limits, replication factors) are assigned via code at VM deployment without manual SAN re-configuration.

3. Network Virtualization (Software-Defined Networking - SDN):

- **Traditional:** Modifying network top-of-rack switches, trunks, access control lists (ACLs), and physical hardware firewalls requires manual intervention and creates change risks.
- **SDN Agility:** The network control plane is decoupled from data plane hardware switches. Logical switches, routers, distributed firewalls, and load balancers are spun up programmatically in software in seconds.

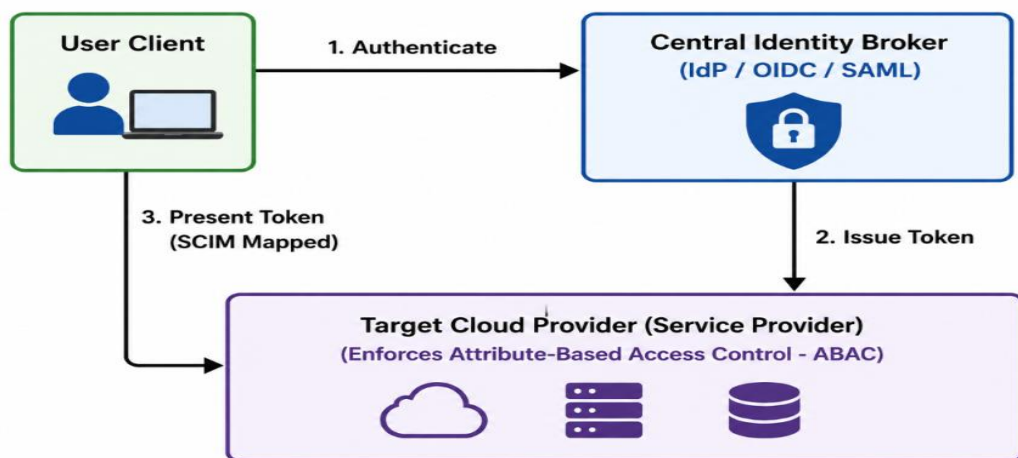
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

When multi-cloud federation projects fail due to identity mismatches, the root causes usually stem from unaligned schema formats, incompatible token formats, and disjointed trust relationships across providers.

Failure Analysis

- **Incompatible Protocols:** Cloud Provider A uses OAuth2/OIDC, while Cloud Provider B relies on legacy SAML or active directory assertions.
- **Attribute Schema Mismatch:** System attributes (e.g., userPrincipalName vs. email or uid) do not align, breaking permission mappings.
- **Missing Mutual Trust & Token Translation:** The target cloud cannot cryptographically validate authorization assertions issued by the origin cloud.

Secure Identity & Privacy Design Framework



1. **Unified Identity Federation Protocols (SAML 2.0 & OIDC):**

- Standardize all enterprise cloud entry points on **OpenID Connect (OIDC)** over **OAuth 2.0** for modern APIs and **SAML 2.0** for enterprise web portals.
- Establish a central **Identity Broker** that acts as the single trusted Identity Provider (IdP) across external cloud environments (AWS, Azure, GCP).

2. **Cross-Domain Schema Standard (SCIM 2.0):**

- Adopt **SCIM 2.0 (System for Cross-domain Identity Management)** to automatically synchronize user accounts, group memberships, and role attributes across clouds using standard JSON payloads.

3. **Attribute-Based Access Control (ABAC):**

- Transition from static Role-Based Access Control (RBAC) to dynamic **ABAC**. Access decisions evaluate environment variables, user risk context, request time, and token attributes rather than hardcoded cloud-specific group names.

4. **Privacy-Preserving Tokens (Pseudonymization):**

- To prevent Personally Identifiable Information (PII) leakage across federated third-party clouds, issue **Pairwise Pseudonym Identifiers (PPIDs)** or transient SAML Subject NameIDs instead of sharing raw employee IDs or email addresses.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear causeeffect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach

Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded: